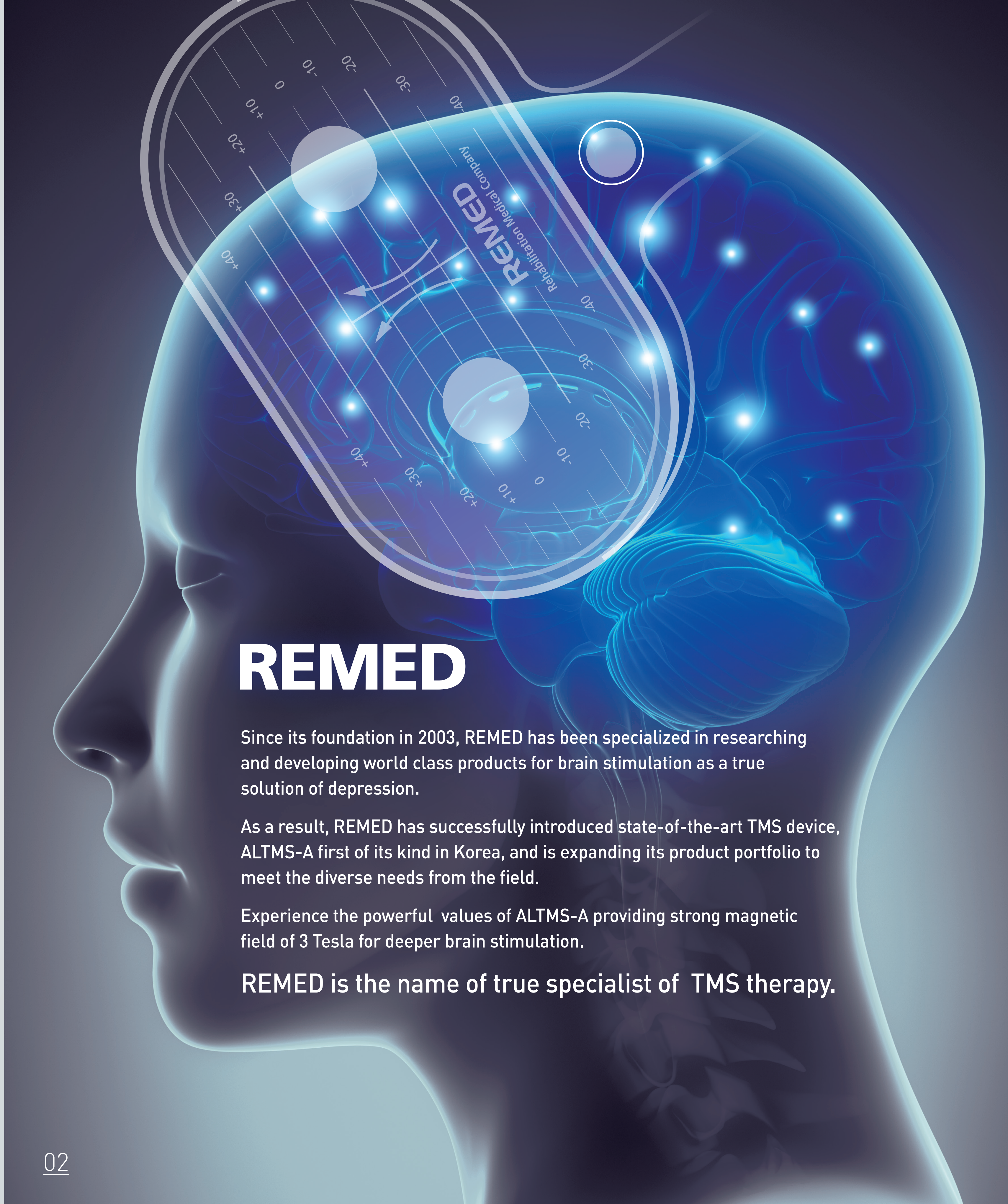
THE TRUE SPECIALIST IN TMS THERAPY

REMED
Rehabilitation Medical Company

WHEN YOUR PATIENTS ARE SUFFERING FROM

REMED PROVIDES TRUE SOLUTION FOR THEM AS GENUINE SPECIALIST OF TMS THERAPY.



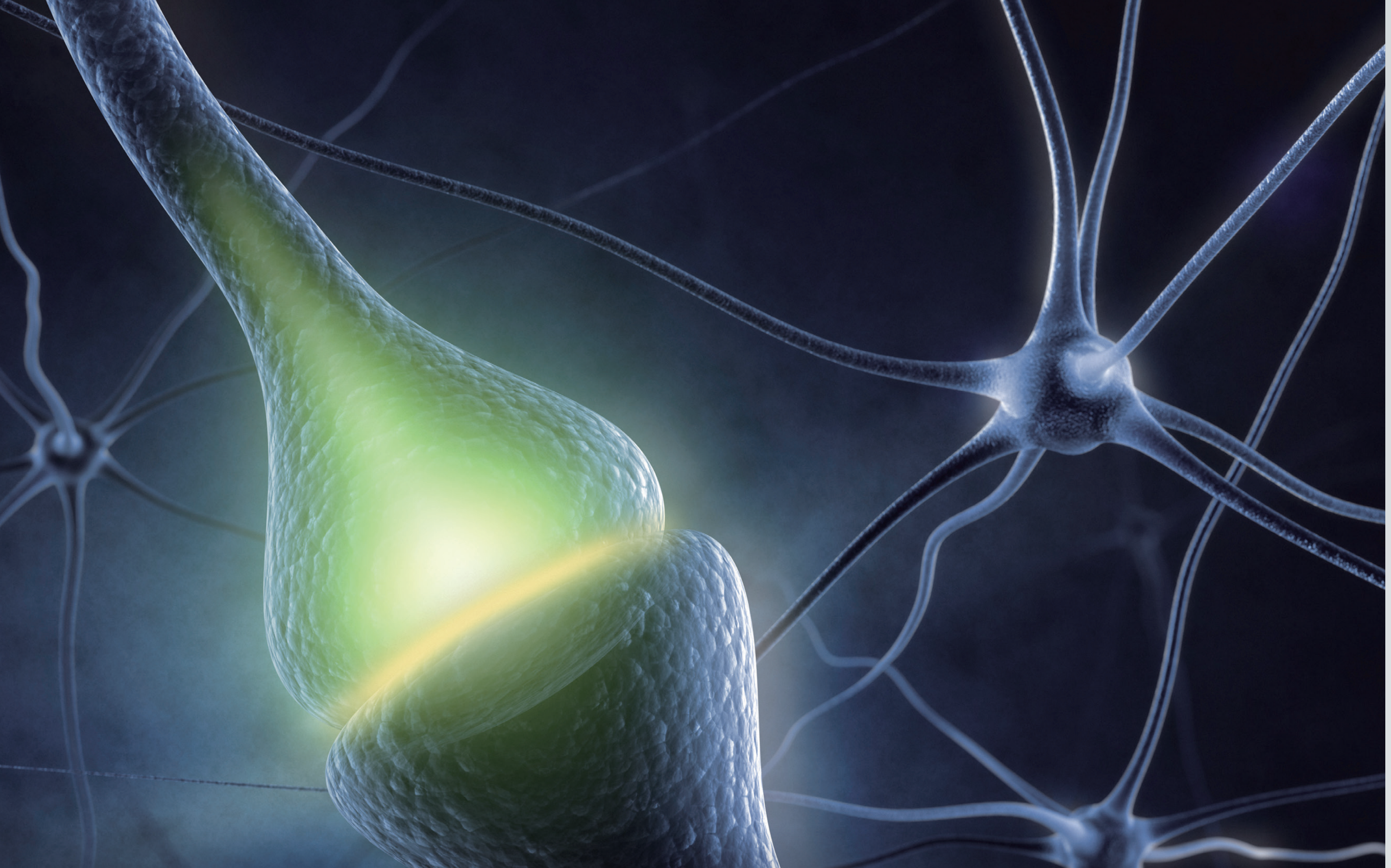
REMED

Since its foundation in 2003, REMED has been specialized in researching and developing world class products for brain stimulation as a true solution of depression.

As a result, REMED has successfully introduced state-of-the-art TMS device, ALTMS-A first of its kind in Korea, and is expanding its product portfolio to meet the diverse needs from the field.

Experience the powerful values of ALTMS-A providing strong magnetic field of 3 Tesla for deeper brain stimulation.

REMED is the name of true specialist of TMS therapy.



WHY TMS THERAPY?

(TRANSCRANIAL MAGNETIC STIMULATION)

Transcranial Magnetic Stimulation(TMS) is a safe, non-invasive therapy method that uses rapidly changing magnetic field to stimulate the neurons in the brain. It is indicated for the treatment of Major Depressive Disorder(MDD), and other certain conditions, for adult patients who have failed to achieve satisfactory improvement from prior antidepressant medication.

Also, TMS is widely accepted as a unique research tool for the investigation of broad variety of issues in different fields:

Cognitive Neuroscience- in studying mental processes related to memory, speech, perception and functional connection

Psychiatry- to influence specific brain function within the dorsolateral prefrontal cortex

Neurophysiology- used in the stimulation of the peripheral and central nerve pathways

Rehabilitation- used in the promotion of muscle recovery and the relief of nerve spasticity

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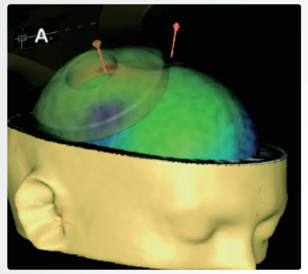
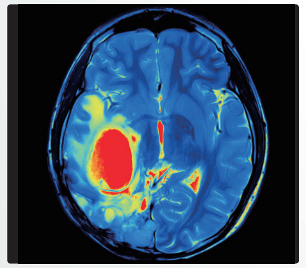
TECHNICALLY PROVEN MECHANISM OF TMS

TMS therapy is an outpatient procedure and does not require anesthesia or sedation, so the patient can go back to daily life right after treatment. Unlike other treatments, such as electro-convulsive therapy(ECT), TMS does not cause cognitive side effects or memory problems.



Time-varying electrical current in a coil is produced.

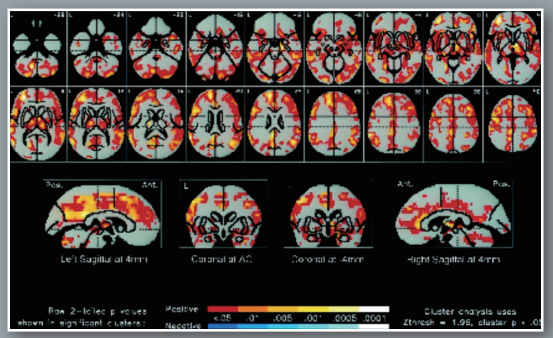
More focused magnetic field from figure of 8 shaped transducer passes unimpeded through skull.



Neurons in the cortex of brain (PFC) is stimulated to lead to behavioral changes.

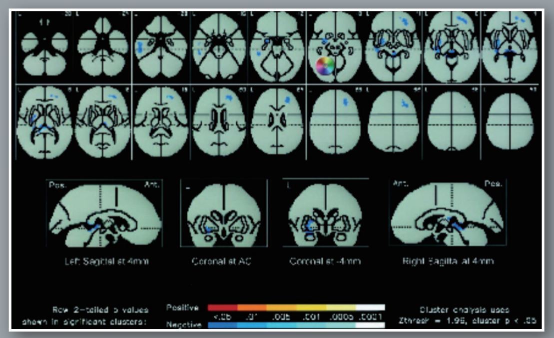
MRI images for patients with Depression and ADHD after TMS treatment

20 Hz



Patient with Depression

1 Hz



Patient with ADHD

The true specialist of TMS for deeper brain stimulation

ALTMS®

Most reliable & powerful TMS Device

Experience the powerful values of ALTMS® providing strong magnetic field of 3 Tesla for deeper brain stimulation.

Features & Benefits

- Convenient & easy operation by touch screen color display
- Unique design of 8 shaped transducer for focused brain stimulation
- Patented oil circulating transducer for intensive & non-tiring treatment
- 2 stimulation modes for diverse applications: Single, Repetitive

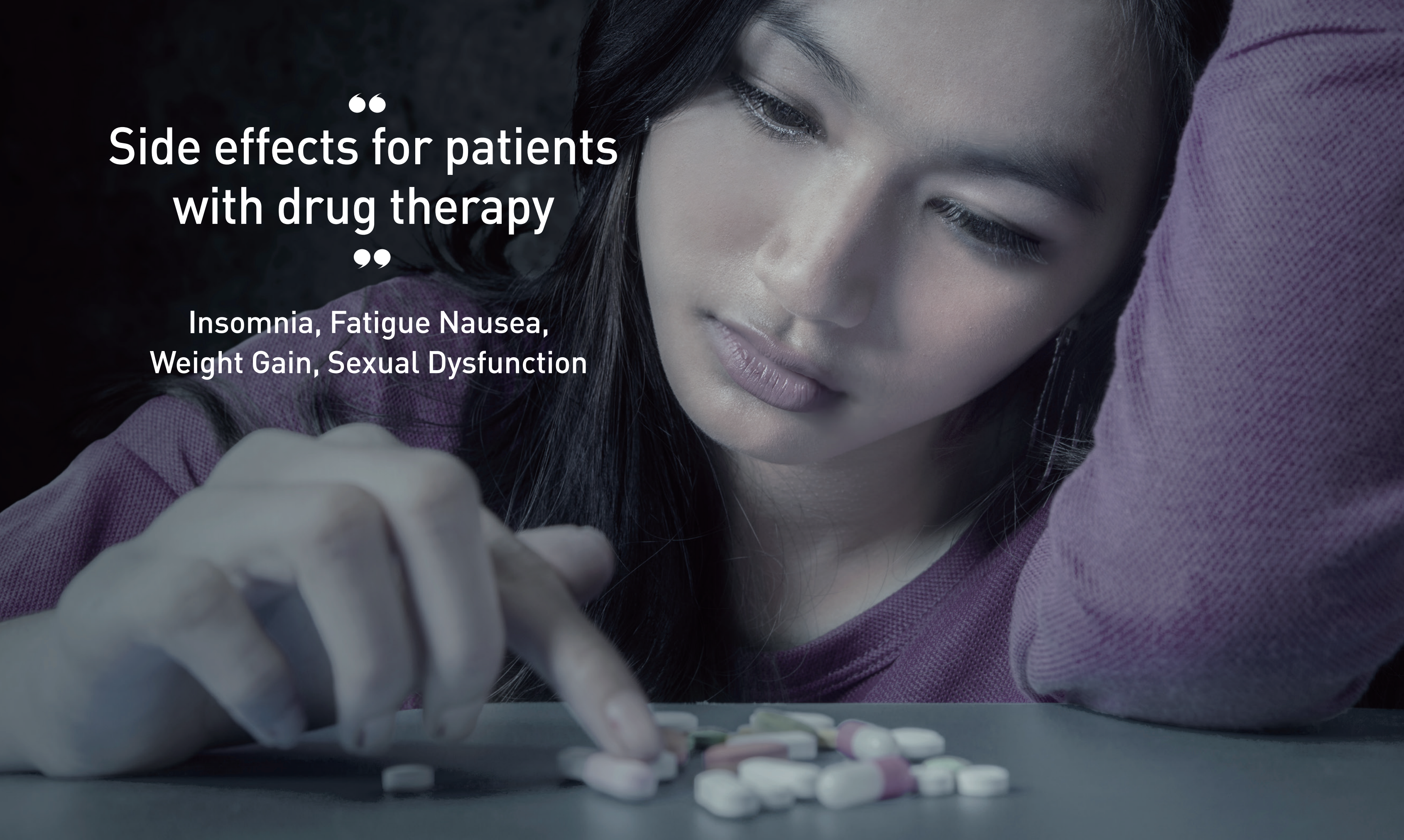
Technical Specifications

Peak Magnetic Energy	1 Tesla peak @ 100% output
Frequency	Maximum 30Hz
	Minimum 0.1Hz
Train Duration	0.1–1Hz: 1–1800 sec, 2–30Hz: 1–20 sec increment
Output waveform	Symmetric biphasic pulse
Weight	61Kg
Power Supply	120V, 60Hz
Software	Integrated color touch screen display
	2 Stimulation mode (Single, Repetitive)

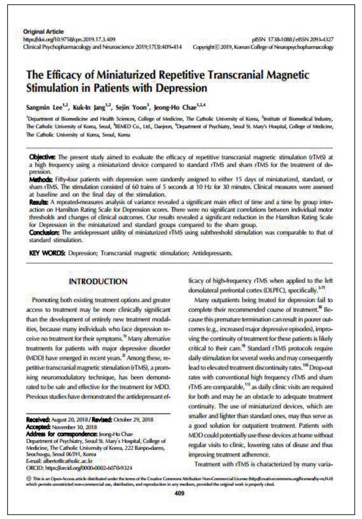
For more detailed clinical indications, please refer to the following reference site:

https://www.ncbi.nlm.nih.gov/pubmed/

ALTMS®



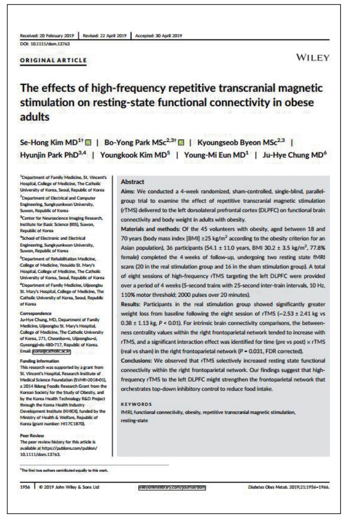
More than thousands of clinical papers available



Original Article
https://doi.org/10.9758/cpn.2019.17.3.409 pISSN 1738-1088 / eISSN 2093-4327
Clinical Psychopharmacology and Neuroscience 2019;17(3):409-414 Copyright© 2019, Korean College of Neuropsychopharmacology

The Efficacy of Miniaturized Repetitive Transcranial Magnetic Stimulation in Patients with Depression

Sangmin Lee[1,2], Kuk-In Jang[1,2], Sejin Yoon[3], Jeong-Ho Chae[1,2,4]



Received: 20 February 2019 | Revised: 22 April 2019 | Accepted: 30 April 2019
DOI: 10.1111/dom.13763

ORIGINAL ARTICLE WILEY

The effects of high-frequency repetitive transcranial magnetic stimulation on resting-state functional connectivity in obese adults

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Abstract

Aims: We conducted a 4-week randomized, sham-controlled, single-blind, parallel-group trial to examine the effect of repetitive transcranial magnetic stimulation (rTMS) delivered to the left dorsolateral prefrontal cortex (DLPFC) on functional brain connectivity and body weight in adults with obesity.

Materials and methods: Of the 45 volunteers with obesity, aged between 18 and 70 years (body mass index [BMI] ≥25 kg/m² according to the obesity criterion for an Asian population), 36 participants (54.1 ± 11.0 years, BMI 30.2 ± 3.5 kg/m², 77.8% female) completed the 4 weeks of follow-up, undergoing two resting state fMRI scans (20 in the real stimulation group and 16 in the sham stimulation group). A total of eight sessions of high-frequency rTMS targeting the left DLPFC were provided over a period of 4 weeks (5-second trains with 25-second inter-train intervals, 10 Hz, 110% motor threshold; 2000 pulses over 20 minutes).

Results: Participants in the real stimulation group showed significantly greater weight loss from baseline following the eight session of rTMS (−2.53 ± 2.41 kg vs 0.38 ± 1.13 kg, P < 0.01). For intrinsic brain connectivity comparisons, the between-ness centrality values within the right frontoparietal network tended to increase with rTMS, and a significant interaction effect was identified for time (pre vs post) x rTMS (real vs sham) in the right frontoparietal network (P = 0.031, FDR corrected).

Conclusions: We observed that rTMS selectively increased resting state functional connectivity within the right frontoparietal network. Our findings suggest that high-frequency rTMS to the left DLPFC might strengthen the frontoparietal network that orchestrates top-down inhibitory control to reduce food intake.

KEYWORDS
fMRI, functional connectivity, obesity, repetitive transcranial magnetic stimulation, resting-state

ARTICLE IN PRESS

Contents lists available at ScienceDirect

Brain Stimulation

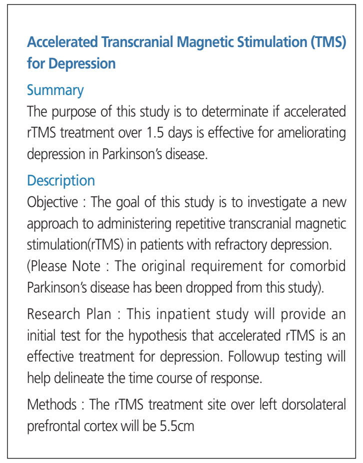
journal homepage: http://www.journals.elsevier.com/brain-stimulation

The effects of repetitive transcranial magnetic stimulation on body weight and food consumption in obese adults: A randomized controlled study

Se-Hong Kim, Ju-hye Chung, Tae-Hong Kim, Seong Hoon Lim, Youngkook Kim, Young-Mi Eun, Yun-Ah Lee

ARTICLE INFO

ABSTRACT



Accelerated Transcranial Magnetic Stimulation (TMS) for Depression

Summary

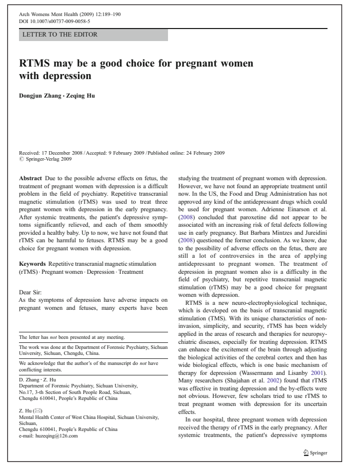
The purpose of this study is to determinate if accelerated rTMS treatment over 1.5 days is effective for ameliorating depression in Parkinson's disease.

Description

Objective : The goal of this study is to investigate a new approach to administering repetitive transcranial magnetic stimulation(rTMS) in patients with refractory depression. (Please Note : The original requirement for comorbid Parkinson's disease has been dropped from this study).

Research Plan : This inpatient study will provide an initial test for the hypothesis that accelerated rTMS is an effective treatment for depression. Followup testing will help delineate the time course of response.

Methods : The rTMS treatment site over left dorsolateral prefrontal cortex will be 5.5cm



Arch Womens Ment Health (2009) 12:189–190
DOI 10.1007/s00737-009-0058-5

LETTER TO THE EDITOR

RTMS may be a good choice for pregnant women with depression

Dongjun Zhang · Zeqing Hu

Received: 17 December 2008 / Accepted: 9 February 2009 / Published online: 24 February 2009
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Ver. 1910.003TMS_ENG